

## Block-aware interferometer synthesis with QROAM-loaded phases

Rows = Cols = 256, the smallest power of two  $\geq 8 \times 26 = 208$ .

Columns/reflections/layers: 256 (rounded up from  $8 \times 26 = 208$  to power of two).

Blocks:  $M = k^3$  for  $k = [1, 2, 3, 4, 5, 6]$ . Phase precision  $b = 32$ .

Interferometer bloq curves use `get_Toffoli_counts / get_qubit_counts` on `BlockUnitaryInterferometerSynthesisQROAM` (the actual Qualtran circuit).

Key differences from theory formula:

- Controlled `AddIntoPhaseGrad` ( $\sim 2\times$  vs uncontrolled) in each beamsplitter layer
- Shifts modeled by  $N//2$  pairs of `AddK` (cyclic increment) instead of analytic formula
- QROAM cost from Qualtran's internal model (may differ by small constants)

Theory formula curves use the  $N \times N$  model from `model_resource_counts.py`:

$T_{\text{layer}} = \text{ceil}(M \cdot N / (2 \cdot L_1)) + 2 \cdot b \cdot L_1 - 5$  (uncontrolled `AddIntoPhaseGrad` approx)

$T_{\text{final}} = \text{ceil}(M \cdot N / L_1) + b \cdot L_1 + \text{ceil}(M \cdot N / L_2) + L_2 - 6$

$T_{\text{total}} = C \cdot T_{\text{layer}} + (n-2) \cdot (N-1) + T_{\text{final}}$  [ $n = \log_2(N)$ ,  $C = \text{layers}$ ]

Additional scaled plots multiply all counts by  $1e+05$ ,  
termed multiplied by QPE iteration and subnormalization. Qubits are unscaled.

Optimization convention:

$\text{Lambda}_1$  is used for layer QROAM and final forward/phase shifts.

$\text{Lambda}_2$  is used for final-layer QROAM uncomputation.

The standard Householder baseline keeps its previous scalar early-stop sweep.

Fitted power laws  $y = c \cdot M^\alpha$ :

Interferometer bloq, Toffoli-opt:  $T \alpha = 0.450$ ,  $c = 6.159e+04$ ;  $Q \alpha = 0.382$ ,  $c = 1.648e+02$

Interferometer bloq, qubit-opt:  $T \alpha = 0.882$ ,  $c = 5.619e+04$ ;  $Q \alpha = 0.025$ ,  $c = 1.101e+02$

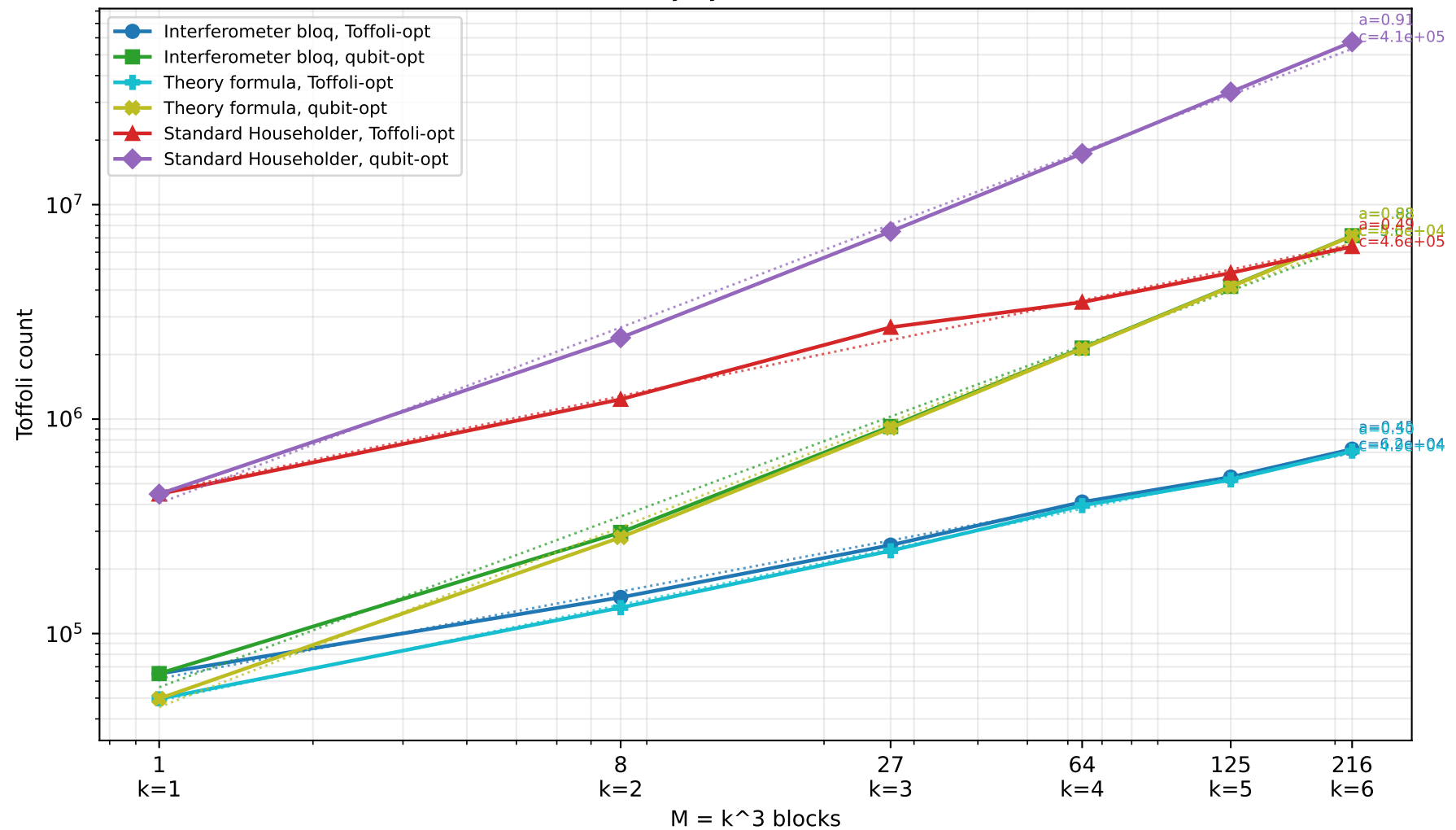
Theory formula, Toffoli-opt:  $T \alpha = 0.496$ ,  $c = 4.856e+04$ ;  $Q \alpha = 0.386$ ,  $c = 1.600e+02$

Theory formula, qubit-opt:  $T \alpha = 0.929$ ,  $c = 4.536e+04$ ;  $Q \alpha = 0.014$ ,  $c = 1.040e+02$

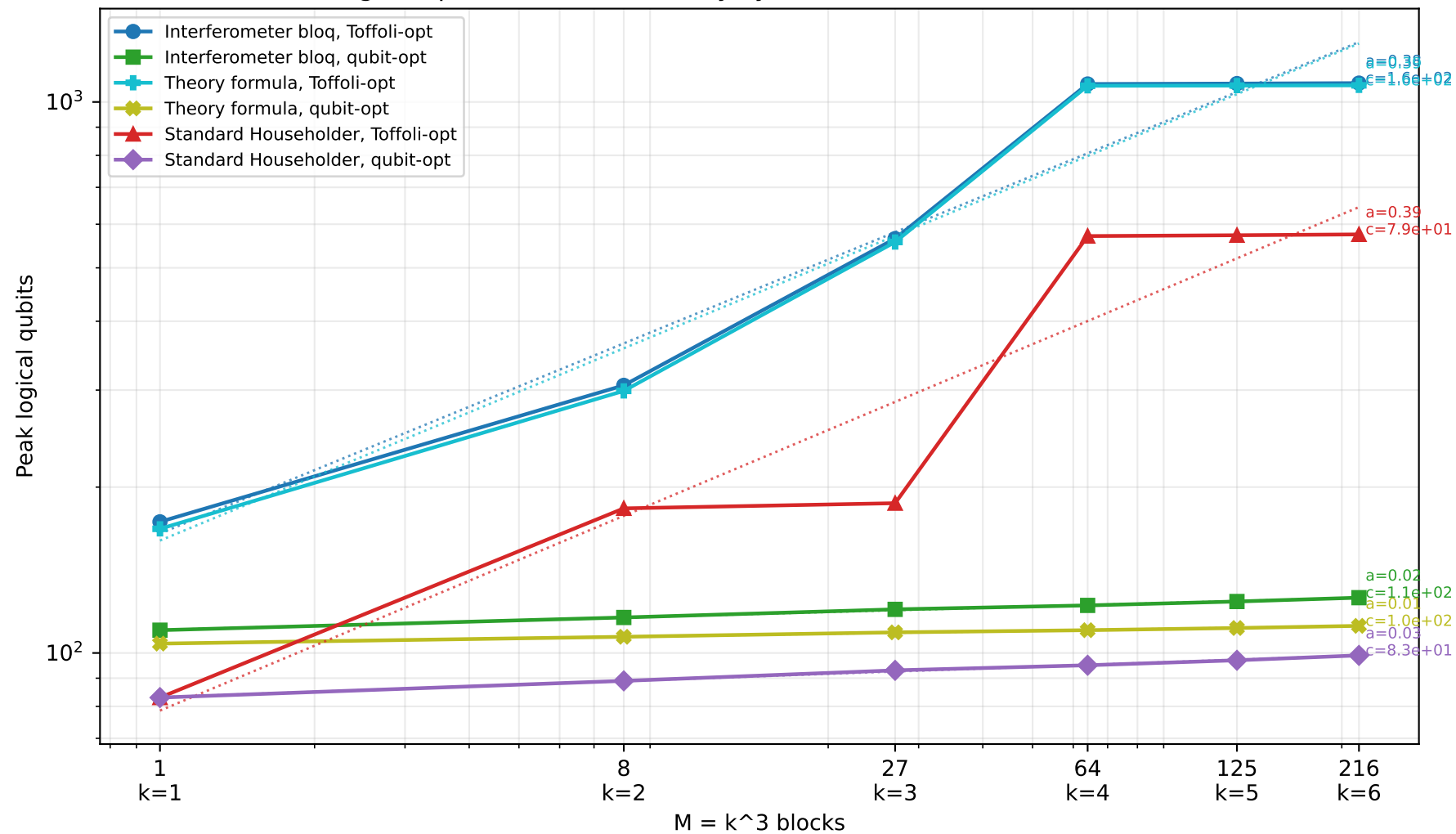
Standard Householder, Toffoli-opt:  $T \alpha = 0.494$ ,  $c = 4.580e+05$ ;  $Q \alpha = 0.391$ ,  $c = 7.860e+01$

Standard Householder, qubit-opt:  $T \alpha = 0.908$ ,  $c = 4.055e+05$ ;  $Q \alpha = 0.032$ ,  $c = 8.314e+01$

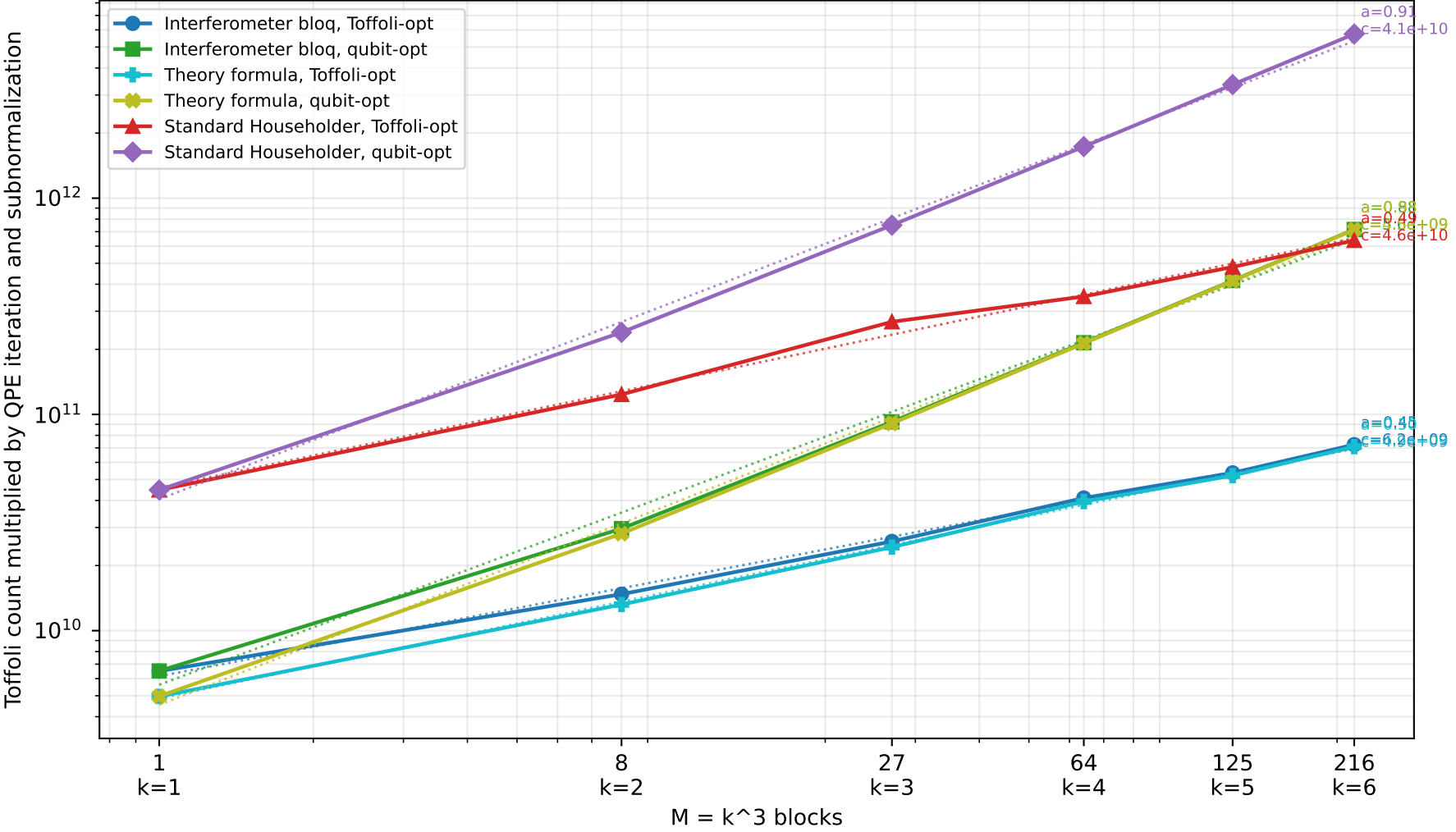
Toffoli count for block unitary synthesis (rows=256, cols=256, b=32)



Peak logical qubits for block unitary synthesis (rows=256, cols=256, b=32)



Toffoli count multiplied by QPE iteration and subnormalization for block unitary synthesis (rows=256, cols=256, b=32); multiplied by QPE iteration and subnormalization



Resource counts and optimization modes

| k | M   | Int T-opt T | Int T-opt Q | Int Q-opt T | Int Q-opt Q | Theory T-opt | Theory T-opt | Theory Q-opt | Theory Q-opt | Std T-opt T | Std T-opt Q | Std Q-opt T | Std Q-opt Q |
|---|-----|-------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|-------------|-------------|-------------|-------------|
| 1 | 1   | 64,984      | 173         | 65,080      | 110         | 49,620       | 168          | 49,716       | 104          | 447,232     | 83          | 447,232     | 83          |
| 2 | 8   | 147,416     | 306         | 296,312     | 116         | 132,052      | 299          | 280,948      | 107          | 1,236,736   | 183         | 2,395,904   | 89          |
| 3 | 27  | 258,564     | 565         | 923,844     | 120         | 243,200      | 557          | 908,480      | 109          | 2,681,088   | 187         | 7,500,544   | 93          |
| 4 | 64  | 410,616     | 1078        | 2,145,816   | 122         | 395,252      | 1070         | 2,130,516    | 110          | 3,510,016   | 571         | 17,313,536  | 95          |
| 5 | 125 | 536,642     | 1080        | 4,160,402   | 124         | 521,278      | 1071         | 4,145,224    | 111          | 4,806,400   | 573         | 33,512,704  | 97          |
| 6 | 216 | 724,560     | 1082        | 7,165,680   | 126         | 709,196      | 1072         | 7,150,772    | 112          | 6,380,288   | 575         | 57,442,048  | 99          |